RAG Learning

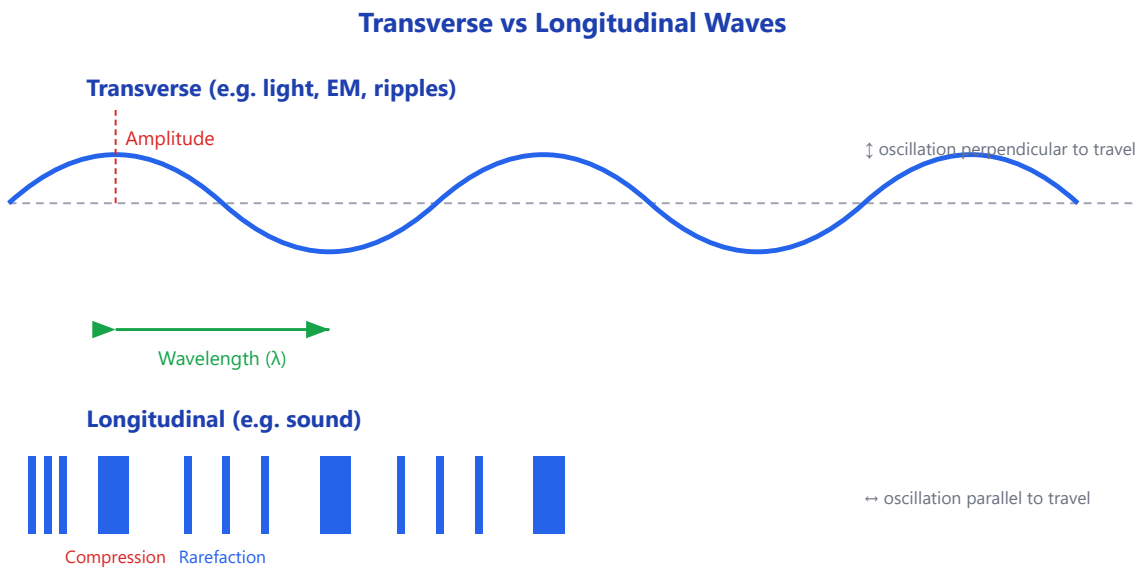
P4 — Waves and Radioactivity

OCR GCSE Physics A (J249) — Gateway Science · Paper 2

P4.1 Wave Basics

A **wave** transfers **energy** (and information) from one place to another *without* transferring matter. The particles of the medium only oscillate about a fixed position — they don't travel with the wave.

Transverse vs Longitudinal Waves



Type	Oscillation direction	Examples
Transverse	Perpendicular to energy transfer	Light, all EM waves, water ripples, string vibrations, S-waves in earthquakes

Longitudinal	Parallel to energy transfer (compressions and rarefactions)	Sound, P-waves in earthquakes, ultrasound

Wave Quantities

- **Wavelength (λ)** — distance between two equivalent points (e.g. crest to crest). Unit: m.
- **Amplitude (A)** — maximum displacement from the rest position. Unit: m. Bigger amplitude = more energy.
- **Frequency (f)** — number of complete waves passing a point per second. Unit: hertz (Hz).
- **Period (T)** — time for one complete wave to pass. T = 1 / f. Unit: s.

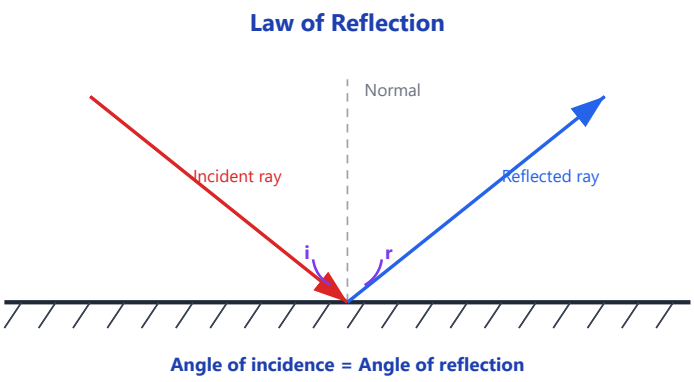
$$\mathbf{v = f \times \lambda} \quad (\text{m/s}) = (\text{Hz}) \times (\text{m}) \text{ — the wave equation}$$

$$\mathbf{T = 1 / f} \quad (\text{s})$$

Worked Example: A sound wave has frequency 500 Hz and wavelength 0.66 m. Speed?
v = f × λ = 500 × 0.66 = **330 m/s** (speed of sound in air at room temperature).

P4.2 Reflection, Refraction and Diffraction

Reflection



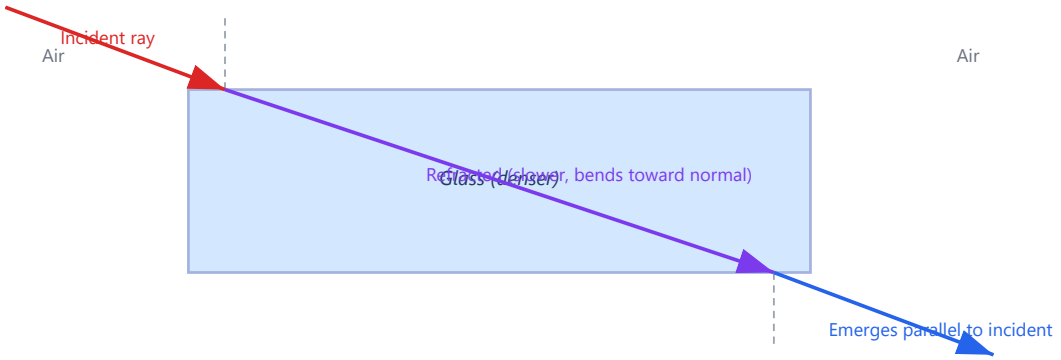
- **Specular reflection** — from a smooth surface (mirror). Parallel rays stay parallel; clear image.

- **Diffuse reflection** — from a rough surface (paper, wall). Rays scatter in many directions; no image.

Refraction

When a wave crosses a boundary between materials of different densities, its **speed changes**. If the wave hits the boundary at an angle, it also changes direction — this is refraction. Wavelength changes but **frequency stays the same**.

Refraction Through a Glass Block



Entering denser medium → bends TOWARDS normal. Leaving → bends AWAY.

> **Exam Tip:** When learning refraction directions: **SLOWER = TOWARDS normal**. A light ray entering a denser medium slows down and bends towards the normal; leaving it speeds up and bends away.

Diffraction

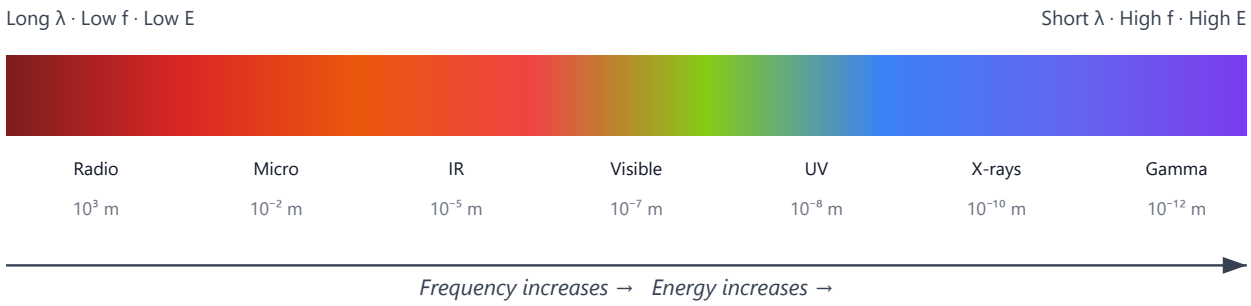
When waves pass through a gap or around an obstacle they spread out — this is **diffraction**. The amount of diffraction is largest when the **gap is similar in size to the wavelength**.

Diffraction — Effect of Gap Size

P4.3 The Electromagnetic (EM) Spectrum

The **electromagnetic spectrum** is a continuous range of transverse waves that all travel at the **same speed** in a vacuum — **3×10^8 m/s (the speed of light)**. They differ only in wavelength and frequency.

The Electromagnetic Spectrum



Memory: Radio Microwave IR Visible UV X-ray Gamma ("Raging Martians Invaded Venus Using X-ray Guns")

Uses and Dangers of EM Waves

Wave	Uses	Dangers
Radio	Broadcasting (radio, TV), communications	Very low energy — negligible hazard
Microwave	Cooking food (absorbed by water); satellite communications, phone signals	Can heat body tissues internally at high doses
Infrared	Electric heaters, TV remotes, thermal imaging, fibre-optic links	Burns from high-intensity IR (causes skin/retina damage)
Visible light	Seeing, cameras, photography, fibre-optic communications	Intense light can damage retina
Ultraviolet	Fluorescent lamps, tanning, security marks, sterilisation	Sunburn, skin cancer, eye damage

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X-rays	Medical imaging of bones, airport security, industrial inspection	Ionising — causes mutations, cancer. Dose must be controlled.
Gamma	Sterilising medical instruments, cancer treatment (radiotherapy), tracers	Most ionising — strong cancer risk. Shielding required (lead).

Exam Tip: Dangers increase with **frequency**. UV, X-rays and gamma rays are **ionising** — they carry enough energy per photon to knock electrons out of atoms, which damages DNA and can cause cancer.

Absorbed / Reflected / Transmitted

When an EM wave hits a material, it can be **reflected**, **absorbed**, or **transmitted**. Which happens depends on the material and the wave's frequency:

- A black surface absorbs visible light & IR — that's why it gets hot in the sun.
- A silver surface reflects IR — hence foil survival blankets reflect body heat back.
- Glass transmits visible light but absorbs UV and IR — so a greenhouse warms up.

P4.4 Sound and Ultrasound

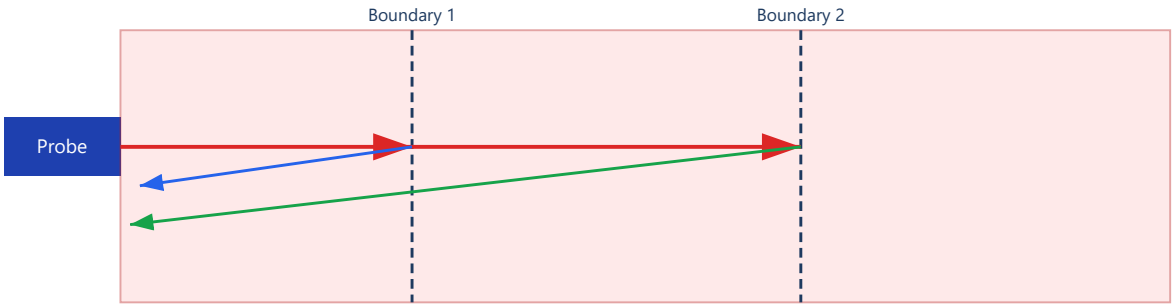
Sound waves are **longitudinal** and need a medium (solid, liquid, gas) — they cannot travel through a vacuum. In air at room temperature, sound travels at about **330 m/s**; faster in liquids (~1500 m/s in water); fastest in solids (5000 m/s in steel).

Hearing Range

- Humans hear roughly **20 Hz – 20 000 Hz (20 kHz)**.
- **Ultrasound** = above 20 kHz. Used for pre-natal scanning, cleaning jewellery, echo-location, non-destructive testing (finding cracks in metal).
- **Infrasound** = below 20 Hz. Used for monitoring earthquakes, volcanoes, whale calls.

Ultrasound Imaging

Ultrasound Imaging (partial reflection at boundaries)



Echo delay → depth of each boundary. Combined echoes build the image.

Ultrasound pulses partially reflect at **each boundary** between different tissues. By timing the echoes and using **d = v × t** (halved — there and back), the depth of each boundary is found. Safer than X-rays for pregnancy scans because it is *non-ionising*.

P4.5 The Atomic Model

History of the Atom

Model	Who	Key idea
Solid-sphere	Dalton (1803)	Atoms are tiny indestructible solid spheres, one type per element.
Plum pudding	Thomson (1897)	Positively-charged "pudding" with electrons ("plums") embedded in it.
Nuclear	Rutherford (1911)	*Gold-foil experiment*: most α-particles passed straight through, a few deflected. Most of the atom is **empty space**; mass is concentrated in a tiny, positive **nucleus**.
Planetary	Bohr (1913)	Electrons orbit the nucleus at fixed energy levels (shells).
Nuclear (with neutrons)	Chadwick (1932)	Discovered the neutron — explained why atoms were heavier than their proton count suggested.

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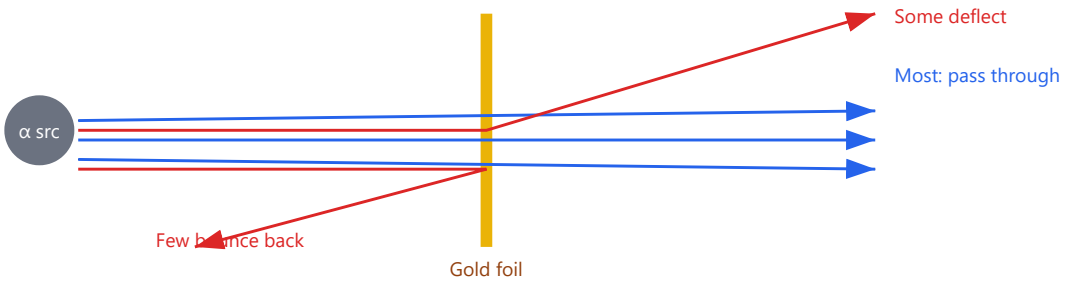
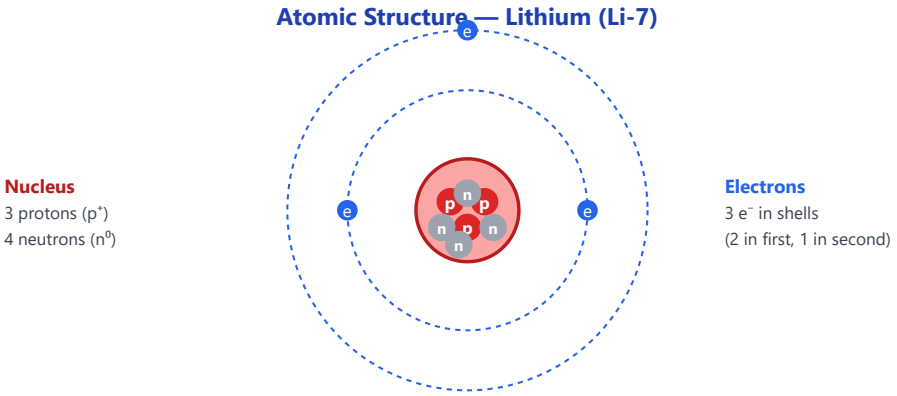
Rutherford's Gold-Foil Experiment



Conclusions from Rutherford's experiment: atoms are **mostly empty space**; they have a **small, dense, positively-charged nucleus**; electrons orbit that nucleus.

Atomic Structure

Atomic Structure — Lithium (Li-7)



Particle	Charge (relative)	Mass (relative)	Location
Proton (p^+)	+1	1	Nucleus
Neutron (n^0)	0	1	Nucleus
Electron (e^-)	−1	1/1836 (≈ 0)	Shells around nucleus

- **Atomic (proton) number Z** — number of protons. Defines the element.
- **Mass number A** — protons + neutrons.
- **Isotopes** — atoms of the same element (same Z) but different numbers of neutrons (different A). Example: carbon-12 and carbon-14.

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Size scale: atom ≈ 10^{-10} m; nucleus ≈ 10^{-14} m. The nucleus is about **10 000 times smaller** than the atom, yet contains >99.9% of the mass.

P4.6 Radioactive Decay

> An unstable nucleus can lose energy and become more stable by emitting **radiation**. This is **radioactive decay**, and it is a **random process** — we cannot predict when any particular atom will decay.

Types of Nuclear Radiation

Type	What is it	Charge	Ionising power	Range in air	Stopped by
α (alpha)	Helium nucleus (2 p, 2 n)	+2	Very high	A few cm	Paper, skin
β (beta)	High-speed electron from nucleus	−1	Medium	~1 m	Few mm aluminium
γ (gamma)	High-energy EM wave (photon)	0	Low	Hundreds of m	Thick lead / concrete
n (neutron)	Neutron emitted from nucleus	0	Low direct; induces activity	Long	Water / hydrogen-rich shielding

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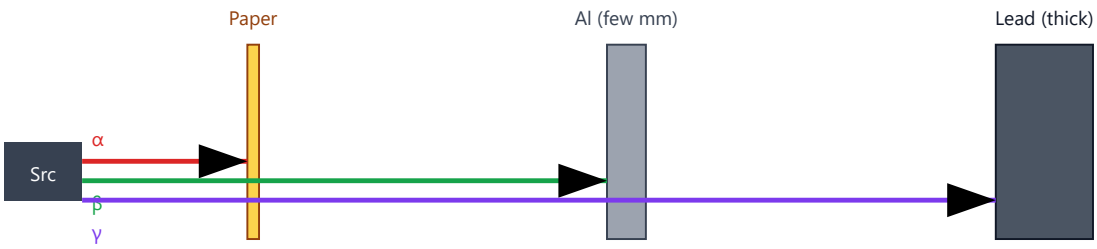
Penetration of α, β and γ



Nuclear Equations

In a decay equation: the mass number (top) and atomic number (bottom) must **balance** on both sides.

$$\alpha \text{ decay: } {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

$$\text{Example: } {}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

$$\beta^{-} \text{ decay: a neutron turns into a proton + electron. } {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e$$

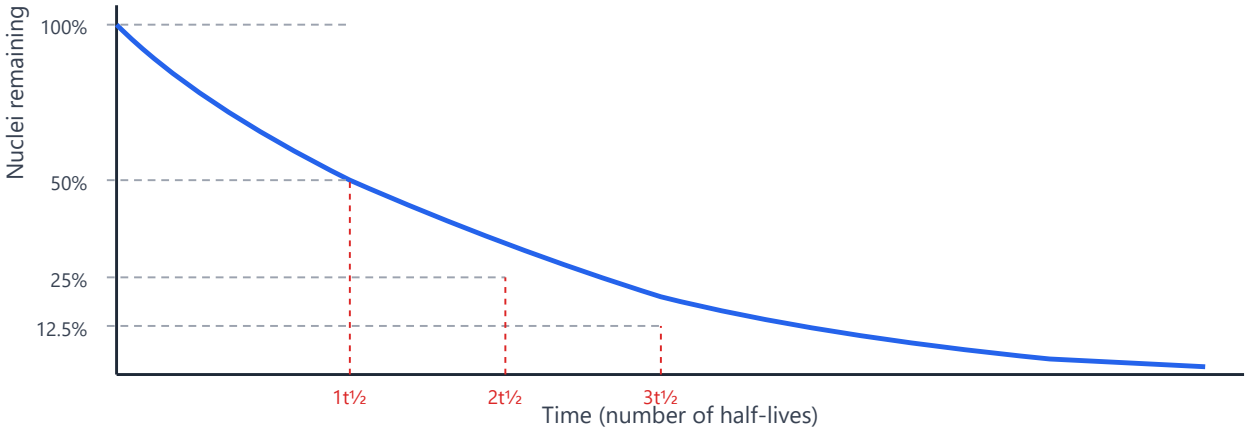
$$\text{Example: } {}^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$$

Gamma emission is different — a γ ray carries off excess *energy* without changing proton or neutron count.

Half-Life

> The **half-life** is the average time taken for the number of undecayed nuclei (or the activity) of a radioactive sample to halve. It is constant for any given isotope.

Radioactive Decay — Half-Life Curve



Worked Example: A sample has an activity of 800 Bq and a half-life of 15 minutes. What will its activity be after 1 hour?
1 hour = 4 × 15 minutes = 4 half-lives.
800 → 400 → 200 → 100 → **50 Bq**.

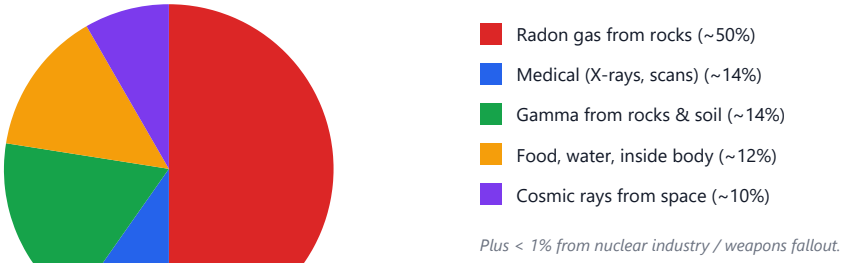
Irradiation vs Contamination

	Irradiation	Contamination
What happens	Object is exposed to radiation from an external source	Radioactive material gets onto / into the object
Continued hazard?	Stops when source removed / shielded	Continues as long as contamination remains
Example	X-ray at the hospital	Radioactive dust after a nuclear accident
Protection	Distance, shielding, time	Clean clothing, gloves, masks, decontamination

Background Radiation

We are all exposed to low-level **background radiation** from natural and artificial sources.

Typical Sources of UK Background Radiation

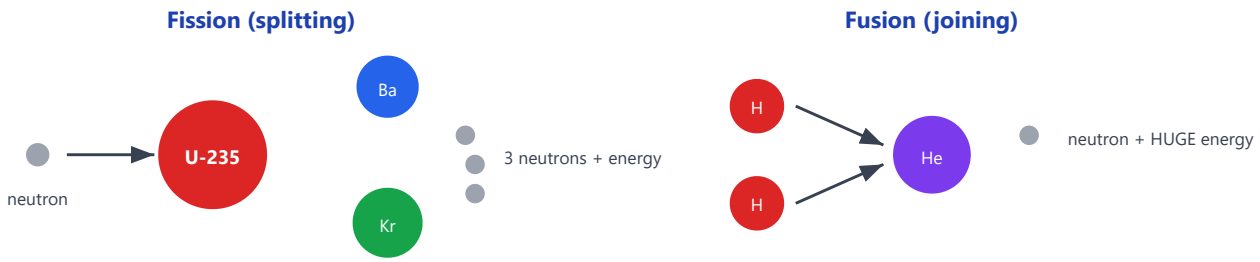


Uses of Radioactivity

- **Medical tracers** — short-half-life γ-emitters (e.g. Tc-99m) injected and tracked to image organs.
- **Radiotherapy** — high-dose γ or X-rays aimed at tumour cells.
- **Sterilising medical equipment** and food — γ rays kill microbes without physical contact.
- **Smoke alarms** — small α source ionises air; smoke disrupts the current, triggering alarm.
- **Industrial thickness control** — β source + detector give consistent paper/metal thickness.
- **Radiocarbon dating** — C-14 ratio in organic material tells you how long ago it died.

Nuclear Fission and Fusion

Nuclear Fission vs Fusion



	Fission	Fusion
Process	A heavy nucleus (U-235, Pu-239) splits into two smaller nuclei + neutrons + energy	Two light nuclei (H isotopes) join to make a heavier nucleus + neutron + energy

Conditions	A slow neutron absorbed by fissile nucleus	Extremely high temperature & pressure (fuses hydrogen in stars)
Chain reaction	Yes — neutrons from each split trigger more splits	No (on Earth; stars sustain it via gravity)
Current use	Nuclear power stations, nuclear weapons	In stars; experimental reactors (ITER, JET). Still research-stage on Earth.
Waste	Long-lived radioactive fission products	Helium (stable, harmless)

Quick Summary — Key Equations for P4

Quantity	Equation	Units
Wave equation	$v = f \lambda$	m/s
Period	$T = 1 / f$	s
Distance from echo	$d = v t / 2$	m
α decay	$A \rightarrow A-4$, $Z \rightarrow Z-2$	—
β^- decay	A same, $Z \rightarrow Z+1$	—
γ decay	A and Z unchanged	—

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